

Alwin Tay Jing Xue

Machine Learning Engineer (LLM)

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Portfolio : www.alwint.dev
GitHub : github.com/hmmm2

PROFESSIONAL SUMMARY

Machine learning engineer with applied experience across LLM fine-tuning, retrieval systems, distributed data pipelines, and multi-tenant backend infrastructure. Built and measured a QLoRA fine-tuning pipeline, a 2.3 million row PySpark feature pipeline, and a security-hardened agentic document platform.

SKILLS

- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, QLoRA, LoRA, PEFT, EfficientNet, DeiT, YOLOv8
- **LLM & Retrieval:** RAG, reranking, embeddings, Qdrant, pgvector, LangChain, prompt engineering, vLLM, Ollama
- **Data & Big Data:** Python, SQL, PySpark, Apache Spark, ETL, pandas, NumPy, time series analysis
- **Backend & MLOps:** FastAPI, PostgreSQL, Redis, Docker, AWS, GitHub Actions CI/CD
- **Languages:** Python, R, Java, C++, SQL, TypeScript, JavaScript
- **Spoken Languages:** English, Chinese (Mandarin), Bahasa Melayu

PROFESSIONAL EXPERIENCE

LLM Research Assistant (1 Year Contract)

December 2025 - Present

Swinburne University of Technology (Sarawak Campus), Sarawak, Malaysia

- Trained a **QLoRA adapter** at **rank 16** with 40.4 million **trainable parameters**, 0.92 percent of **Qwen2.5 7B-Instruct**, over 1,860 curated instruction examples on a single 16 GB **RTX 5060 Ti** at effective **batch 32** and **sequence length 1,536**.
- Reduced the training footprint to a measured 10.4 GiB peak against roughly 112 GiB for **full-parameter AdamW**, through **4-bit NF4 quantization** with **rank-16 LoRA adapters**.
- Evaluated the tutor across 9 runs on its **production serving path**, reaching **bug-localisation accuracy of 0.813** with a **95% confidence interval** of 0.774 to 0.846, **routing accuracy** of 0.941, and a **debug-mode recall** of 0.902.
- Engineered a **streaming Python FastAPI backend** with a **Next.js** and **TypeScript** client over **PostgreSQL** for a document reasoning application.

LLM Research Intern

June 2025 - December 2025

Swinburne University of Technology (Sarawak Campus), Sarawak, Malaysia

- Implemented a **Low-Rank Adaptation** bias-detection workflow orchestrated in **LangChain** with **Ollama inference**, evaluating **Llama 3 8B** against **Direct Preference Optimization (DPO)** baselines with **Qdrant vector retrieval**.
- Authored a research paper on a recursive **refinement and bias analysis framework for conversational AI** under academic supervision.

Software Business Analyst (Full-time)

Nov 2023 - June 2024

Finology Sdn Bhd, Kuala Lumpur, Malaysia

- **Analyse business requirements** and **translate them into 40+ technical Jira specifications**, defining edge cases and data validation rules to reduce developer rework by 30% across **two fintech projects (Aeon Bank & AHAM Asset Management)**.
- **Executed technical quality assurance (SIT/UAT) across staging and production environments**, isolating functional bugs in loan application workflows before deployment.
- Collaborated in an **Agile Scrum team** throughout the **SDLC**, owning **Functional Specification Document (FSD)** updates for client sign-off and applying knowledge of the software architecture to align specifications with system design.

AI/Machine Learning Intern

Oct 2023 - Nov 2023

Hexabyn Technologies Sdn. Bhd, Sarawak, Malaysia

- Develop a **Fruit Defect Detection System** using **EfficientNet-B0**, achieving **93.4% classification accuracy**.
- Curated the annotated image dataset and implemented an iterative **PyTorch training pipeline** to accelerate **model optimization**.
- Developed a **Django** interface to integrate **YOLOv8** output into a functional **quality-inspection workflow**.

Quality Control Engineer Intern (IT Programmer)

Mar 2023 - Jul 2023

LONGi Malaysia Sdn. Bhd, Sarawak, Malaysia

- Engineered a **VB.NET and MySQL automation system** for solar cell certification, processing high-volume logic to **reduce manual generation by 85% and boost engineering productivity by 60%**.
- Developed automated **Python ETL pipelines for 22 real-time datasets** and conducted **time series analysis**, deriving **data-driven insights to predict and mitigate manufacturing defects**.

TECHNICAL PROJECTS

SYNTHIEN AI (Multi-Tenant Document Intelligence Platform)

2025 - Present

- Built an **agentic retrieval backend** across 197 source files and 79 migrations with 9 registered capabilities, **hybrid search** and **reranking** with per-claim citations, on **FastAPI** with **PostgreSQL**, **Qdrant** and **Redis**.
- Enforced fail-closed **row-level security** across 56 policies at the data layer, verified by an 85-test **adversarial suite** covering **cross-tenant reads**, **filter injection**, **prompt injection** and redaction before **model egress**.
- Instrumented **PII redaction** before every **outbound model call** and an append-only audit log spanning 88 action types, with **continuous integration** running lint, type-check, contract-drift and a dedicated tenancy job on Linux service containers.

VoidCode AI (ML Interview Preparation Platform)

2025 - Present

- Built a **PySpark feature pipeline** processing 2.3 million code-submission rows at roughly 9,200 rows per second on 10 cores into **per-learner mastery vectors** for 117,000 learners.
- Trained a **LambdaMART ranker** over those features, reaching **Recall at 100** of 0.616 against a **0.190 popularity baseline** and an **NDCG at 10** of 0.214.
- Authored and **validated a 200-item catalogue** with **125 executable problems** and **699 test cases**, every reference solution executing green in continuous integration.

EDUCATION

Swinburne University Of Technology

September 2024 - June 2026

Bachelor of Computer Science (Major in Data Science)

LICENSE & CERTIFICATION

- PyTorch for Deep Learning Professional Certificate by DeepLearning.AI [\[link\]](#)
- TensorFlow Developer Professional Certificate by DeepLearning.AI [\[link\]](#)
- AI Engineer for Developers Associate Certificate by DataCamp [\[link\]](#)
- AI Engineer for Data Scientists Associate Certificate by DataCamp [\[link\]](#)
- Data Scientist Associate Certificate by DataCamp [\[link\]](#)
- Product Management Certification by General Assembly [\[link\]](#)
- Statistical Analysis System (SAS) Certificate [\[link\]](#)

AWARDS & HONORS

- **[National]** TechFrontier Explorer × SCENT Demo Day Competition – 2nd Place Award [\[link\]](#)
- **[National]** TechFrontier Explorer × SCENT Programme – Grants Award [\[link\]](#)